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RESEARCH ARTICLE

An Adaptive Convolutional Neural Network With Spatio-Temporal Attention and Dynamic Pathways (ACNN-STADP) for Robust EEG-Based Motor Imagery Classification

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ABSTRACT Electroencephalogram (EEG)-based Brain-Computer Interfaces (BCIs) have gained substantial attention, particularly for motor imagery (MI) that facilitates direct brain-to-device communication without any muscular movement. However, existing classification models face limitations such as inter-subject variability, lack of generalizability, high computational demands, low signal-to-noise ratios, and inefficient feature extraction, which impede their robustness and accuracy. Moreover, advanced deep learning models often utilize rigid architectures with fixed spatial-temporal filters, restricting their adaptability to dynamic EEG patterns. To address these challenges, this paper proposes an Adaptive Convolutional Neural Network with Spatio-Temporal Attention and Dynamic Pathways (ACNN-STADP), which introduces a novel dynamic pathway mechanism and adaptive attention strategy for robust MI-EEG decoding. The proposed model integrates a Dynamic Pathway Convolution Network (DPCN) for adaptive feature extraction, incorporating a Dynamic Gating Controller (DGC) and Dynamic Adaptive Spatio-Temporal (DAST) blocks to efficiently capture multi-scale spatial and temporal dependencies. Additionally, an Adaptive Attention Fusion (AAF) module employs Dual Multi-Head Self-Attention (DMHSA) and a U-Net-inspired Adaptive Fusion Block (AFB) to enhance feature integration and improve classification performance. Furthermore, the model introduces three key innovations: Dynamic Multi-Scale Convolutional Learning for adaptive kernel selection, Unified Spatio-Temporal Attention (USTA) for efficient feature recalibration, and AFB for multi-scale feature fusion while preserving long-range dependencies. The model is validated on BCI Competition IV Dataset 2a, achieving a peak accuracy of 90.77%, and further evaluated across six additional MI-EEG datasets, demonstrating an overall average accuracy above 78.98%. ACNN-STADP significantly improves generalization, reduces computational complexity, and enhances real-time applicability, establishing a robust multi-dataset adaptive deep learning framework for EEG-based MI classification.

INDEX TERMS Brain-computer interface (BCI), electroencephalogram (EEG), motor imagery (MI), classification, adaptive CNN, unified spatio-temporal attention, dynamic pathways convolution.

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I. INTRODUCTION

The brain-computer interface (BCI) allows an individual to control external devices and interact with the environment by decoding neurophysiological signals into actionable

commands. This technology is particularly transformative for people with severe motor impairment, offering an alternative route of communication and control in the event of loss of traditional neuromuscular channels. The brain signal can be recorded using various techniques, such as electroencephalography (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), near-infrared spectroscopy (NIRS), and more invasive methods like electrocorticography (ECoG). Among the different methods used in the BCIs, EEG signals have attracted considerable attention because of their high resolution, ease of use, low cost, low risk and cost-efficiency, which make them suitable for industrial applications. E.E.G. is a noninvasive method of recording electrical activity in the brain [1].

EEG signals are captured on the scalp as a two-dimensional (2-D) matrix of real values of time and channel. EEG-based BCIs leverage distinct brainwave patterns, particularly in the mu (8–12 Hz) and beta (13–30 Hz) frequency bands, which are generated during Motor Imagery (MI) tasks [2]. MI is the activity of thinking about moving a part of the human body without physically moving it. MI-EEG activities have been employed in a variety of medical applications, including stroke rehabilitation, wheelchair control, prostheses control, exoskeleton control, cursor control, spelling, and thought-to-text conversion. MI-EEG signals have also been used in nonmedical applications such as vehicle control, drone control, smart home, security, gaming, and virtual reality.

The decoding of MI-EEG signals poses considerable challenges because EEG signals are inherently non-stationary, with substantial variability across trials and subjects. Furthermore, these signals are often contaminated by environmental noise, artefacts, and physiological interference, which exacerbate the difficulty of accurate decoding [1]. To address these challenges, Machine Learning (ML) techniques have been extensively employed due to their ability to adaptively model the complexities of EEG data. Despite their success, the effectiveness of ML classifiers varies significantly depending on the characteristics of the dataset, the task, and the model architecture. Selecting an optimal classifier remains a formidable challenge, requiring a nuanced understanding of the dataset and meticulous optimization to achieve reliable performance.

Early traditional ML approaches for MI-EEG classification relied on handcrafted feature extraction and shallow classifiers. Common Spatial Patterns (CSP) [3], [4] and Filter Bank CSP (FBCSP) [5] were widely adopted for spatial-frequency feature extraction, often paired with classifiers like SVM [6], LDA [7], k-NN [8], EEGNet [9]. While these methods achieved moderate success, they suffered from labor-intensive feature engineering, sensitivity to noise, and poor generalization across subjects due to EEG non-stationarity [10], [11]. For instance, the research [12] reported only 71% subject-independent accuracy using stacked autoencoders, underscoring the limitations of manual feature selection.

MI-EEG classification achieved a major breakthrough through Deep Learning (DL) models particularly Convolutional Neural Networks (CNNs), triggered end-to-end learning operations. CNNs provide excellent capabilities for EEG signal classification because it automatically extracts both time-based and spatial patterns within raw data [13]. CNNs optimize their capacity to discover stacked signal features because they extract detailed patterns from EEG information which other ML approaches might miss. Traditional CNN techniques perform inadequately when applied to EEG signals because they fail to handle specifically EEG difficulties which include patient-dependent variations together with changing patterns and individual trial differences. The need emerges for adaptive approaches to modify CNN feature extraction capability because static architecture models face issues during effective dataset generalization across diverse datasets [14], [15]. The compact CNN named EEGNet demonstrated 68.8% accuracy in detecting movements, but it used fixed filter dimensions which restricted adaptability to vary between subjects [15]. The combination of Temporal Convolutional Networks (TCN) and EEGNet in EEG-TCNet [15] had a fixed structural design which did not include mechanisms to highlight important spatial and temporal features.

Attention mechanisms enhanced feature discrimination by dynamically weighting spatial or temporal regions. LATCNet [10] implemented multi-head self-attention (MSA) together with TCNs [16] for achieving 70.9% accuracy through temporal attention mechanisms that operated with fixed temporal window sizes although their approach produced discrepant results compared to transient neural event patterns. The research [17] incorporated Efficient Channel Attention (ECA) but applied it statically, failing to adapt to dynamic signal characteristics. Multi-branch CNNs, such as MBEEGSE [18] extracted multi-scale features using parallel branches with Squeeze-and-Excitation (SE) blocks [19]. However, their static reduction ratios (e.g., $r = 2$) and fixed hyperparameters limited adaptability to inter-subject variability. Similarly, TCNet-Fusion [20] employed static feature fusion layers, leaving dynamic pathway selection unexplored.

Feature fusion is a process of combining features from various branches or layers to improve model performance by leveraging their complementary nature. In computer vision, feature fusion is an essential technique for boosting performance [21]. In addition, feature fusion methods have been widely used in the medical field, such as cardiovascular disease assessment [22]. Recently, researchers have integrated feature fusion techniques with DL models to decode MI-EEG. For example, the TCNet Fusion model [15] enhances the performance of EEG-TCN model by introducing fusion layers that combine features from different layers to construct rich feature mappings. The research [23] proposed a multi-scale fusion CNN combining an attention mechanism to decode MI-EEG signals. The network extracts multiscale spatio-temporal features from the signal and uses a

TABLE 1. Summary of key literature.

Author	Technique Name	Preprocessing	Classifier	Dataset	MI Tasks
F. Hwaidi, et al. [9]	DAE + CNN	Variational Autoencoder for noise removal	CNN	Physionet	4-class MI
H. Altaheri et al. [11]	ATCNet	No preprocessing	MSA + Temporal Convolution	BCI-IV 2a	4-class MI
R. Buragohain et al. [28]	WST-CNN	Bandpass filtering	Wavelet Scattering Transform and CNN	BCI-IV 2a	4-class MI
Ingolfsson et al. [15]	EEG-TCNet	Bandpass filtering	Temporal Convolutional Network	BCI-IV 2a	4-class MI
Yuxuan Shi et al. [29]	RGN	No preprocessing	Riemannian Geometry Networks	BCI-IV 2a	4-class MI
H. Ahmadi et al. [25]	WS-AIEC	Band-pass filtering	Ensemble of ML classifiers	BNCI2014-002	Multi-dataset MI
Kai Zhou et al. [30]	DB-ATCNet	No preprocessing	Dual-Branch CNN + MSA	PhysioNet, BCI-IV 2a	2-class, 4-class MI
V. Lawhern et al. [31]	EEGNet	Bandpass filtering	Depthwise and Separable CNN	P300, ERN, MRCP, SMR datasets	Multi-paradigm EEG tasks
Yu Song et al. [32]	AWKNet	Channel wise normalization	Adaptive Wavelet Transform Kolmogorov–Arnold Network	BCI-IV 2a	4-class MI
Xicheng Lou et al. [33]	EEG-DBNet	No preprocessing	EEG-DBNet (Dual-Branch Network)	BCI-IV 2a	4-class MI
Xiao-Cong Zhong et al. [34]	EEG-DG	No preprocessing	Multi-Source Domain Generalization Framework	BCI-IV 2a	4-class MI
Ali et al. [35]	ConTraNet	Bandpass filtering	CNN + Transformer	BCI Competition IV-2b, Physionet MI-EEG, Mendeley sEMG datasets	Multi-dataset MI and EMG
Xiong, S et al. [36]	MBCNN-EATCFNet	Zero-mean normalization	Dense Layer and Adaptive Weighting	BCI Competition IV 2a, IV 2b	2-class, 4-class MI
Tian-jian Luo et al. [37]	S-MVTSF	Time-frequency decomposition, zero-mean normalization	Sparse Support Matrix Machine	BCI Competition III 4a, IV 1, IV 2a	2-class, 4-class MI
Jing Jin et al. [38]	MSTFNet	Data augmentation	CNN-based DL model	BCI-IV 2a, BCI-IV 2b,	2-class, 4-class MI
Raza et al. [39]	CNN-LSTM	Bandpass filtering	CNN-LSTM Classifier	BCI Competition IV dataset 1	3-class MI

multiple-feature fusion method to maintain maximum information flow. The limitations include increased computational complexity and longer training times due to the integration of multiple feature extraction layers. Optimal fusion layer design often requires extensive hyperparameter tuning, which can be challenging with limited labelled data [24]. Additionally, the interpretability of models with complex fusion mechanisms may be compromised.

Recent works attempted to fuse spatial and temporal features. CS-CNN [23] combined raw EEG and MRI-derived source space, achieving 96.05% accuracy but requiring anatomical constraints, which restricts applicability in MRI-free scenarios. WS-AIEC [25] used dynamic classifier weighting but depended on CSP features, neglecting end-to-end learning. The research [26] proposed multitask autoencoders with temporal attention but overlooked spatial

dependencies critical for MI. While sliding window strategies [27] mitigated temporal non-stationarity, their reliance on CSP and LDA limited feature diversity.

These limitations point to the key gaps in current research:

1. **Rigid architectures with limited adaptability:** There is no existing model that dynamically modifies its architecture on the basis of input-specific EEG patterns, resulting in degraded performance in heterogeneous data sets. Most state-of-the-art models such as EEGNet [31], ATCNet [11] and EEG-TCNet [15] employ fixed convolutional filters or static attention mechanisms, which cannot dynamically adjust to varying EEG signal characteristics.
2. **Inefficient spatio-temporal feature integration:** There is no end-to-end model that effectively combines spatial and temporal EEG features in an adaptive

and computationally efficient way. Current methods either focus only on spatial features such as CSP-based models [27], losing critical temporal dynamics or prioritize temporal modelling such as TCNs [16] neglecting spatial channel relationships. Moreover, many work use naive fusion techniques such as simple concatenation [21] which fails to optimally combine spatial and temporal features.

3. **Poor generalization across subjects and tasks:** Existing models lack the ability to adapt to individual neurophysiological differences, resulting in inconsistent performance across subjects. Most models such as [10] and [20] achieve high accuracy only on simple hand or foot MI tasks but degrade significantly for complex MI tasks like tongue movement and combined limb MI. Moreover, static architectures struggle with intra-session variability and inter-subject variability, which limit clinical applicability [34].
4. **High computational complexity without proportional gains:** Existing methods trade computational efficiency for marginal performance gains, which hamper their uptake in resource-constrained environments. Many advanced models such as CS-CNN [23], MBEEGSE [18] introduce redundant parallel computations such as fixed multi-branch CNNs with complex fusion layers that increase training time without significant accuracy improvements. Moreover, their heavy parameterization, makes them impractical for real-time BCI applications.
5. **Limited robustness to real-world noise and artifacts:** There is no model that effectively integrates noise resilience into its structure without compromising the integrity of raw signal information of the signal. Most models rely on heavy preprocessing such as independent component analysis (ICA) and manual artifact removal [5], which discard useful data. Moreover, attention mechanisms [17] are applied statically, failing to suppress noise dynamically.

To overcome these limitations, we propose that this research leverages Adaptive CNNs and Spatio-Temporal Attention with Dynamic Pathways (STADP) to address domain-specific MI-EEG data issues. This produced explainable DL model is reliable and comprehensible for decoding MI-EEG brain signals. This model mainly focuses on the adaptive nature so that it can tackle a diverse range of data without any major change in the decoding pipeline. The suggested model includes a three-step adaptive framework for processing MI-EEG data. Firstly, an adaptive CNN packaged with dual-pathways operates as the initial stage to dynamically process signals by identifying the most important spatial-temporal characteristics. Secondly, Feature integration receives an improvement through an adaptive attention mechanism that refines both spatial and temporal dependencies selectively. In last, advanced temporal features emerge through a hierarchical fusion approach which completes adaptive fusion processes in the model from

highlighted content. This lightweight, self-adaptive framework ensures robust MI classification while an adaptive segmentation approach, autonomously identifying the most informative EEG segments and eliminating the need for fixed sliding-window methods, making it ideal for real-time BCI applications.

We used a multi-dataset approach to test the performance of the proposed model across seven different MI-EEG datasets to validate its adaptability. The datasets were used based on large, medium and small data sizes. Achieving consistent classification average accuracy above 78.98% across all datasets demonstrates robustness and generalization capability of proposed model. This adaptability stems from its dynamic feature extraction and learning mechanisms, enabling it to handle variations in signal quality, dataset size, and subject-specific differences without major modifications to the decoding pipeline. While the methodology is explained based on BCI Competition IV Dataset 2a, the strong performance of model across multiple datasets highlights its potential as a generalized, multi-dataset approach for MI-EEG classification.

This proposed architecture is inspired by [11] and [30] and shares the same kernel parameters as the research [11] but models used depthwise convolutions which have limited capacity to capture spatial information from EEG signals, which may result in the loss of spatial features. To avoid this a convolutional mechanism is proposed that dynamically selects and applies kernel sizes to adaptively capture spatial and temporal dependencies. It includes 3×3 , 5×5 , and 3×3 dilated convolutions, allowing the model to extract both fine-grained and coarse features from EEG signals. The proposed model introduces Unified Spatio-Temporal Attention (USTA) a multi-scale adaptive attention mechanism to recalibrate both temporal and spatial (channel) features in a more simplified manner. It assigns dynamic weights to different EEG channels, emphasizing discriminative features related to MI tasks while suppressing irrelevant information. This model does not use a channel attention mechanism such as ECA twice [30] LAMDA [40], SE [19], CBAM [41], and ECA [41]. Although these techniques increase the extraction without reducing the dimensionality, but the risk of overloading the channel is very high. The research [27], [42] uses a traditional sliding window with fixed constraints; this model introduces Lightweight Temporal Convolution (LTC) to adaptively segment EEG signals, enhancing accuracy, efficiency, and interpretability while maintaining a lightweight architecture [43]. Instead, dividing EEG data into equal-length windows, LTC dynamically identifies and processes the most informative temporal regions, ensuring optimal feature extraction [44]. By applying hierarchical convolutions with varying dilation rates (1.0 and 2.0), LTC effectively captures both short- and long-term dependencies without requiring manual segmentation. This approach reduces preprocessing complexity and improves the robustness of EEG-based feature representation [44], [45].

This aims to address the dual challenges of variability and non-stationarity in EEG signals, thereby improving classification performance and reliability across diverse subjects and tasks.

Contributions of this study include:

1. We develop dynamic multi-scale convolutional learning in which the model dynamically selects kernel sizes to optimize spatial-temporal feature extraction, improving robustness to inter-subject variability.
2. We propose Unified Spatio-Temporal Attention (USTA) an adaptive, lightweight multi-scale attention mechanism that synergistically recalibrates spatial (channel-wise) and temporal features. This simplifies computation, reduces overfitting risks, and aligns attention with transient neural events.
3. We propose Lightweight Temporal Convolution (LTC) for Adaptive Segmentation, a lightweight module that autonomously identifies and processes the most informative EEG segments
4. We introduce a U-Net-inspired Adaptive Fusion Block (AFB) module that dynamically integrates multi-scale features, preserving long-range EEG dependencies while eliminating redundancy and improving generalization across diverse MI-EEG datasets.
5. We validate ACNN-STADP across seven diverse MI-EEG datasets. This demonstrates a multi-dataset approach.

Reproducibility Statement- To support reproducibility, the source code, trained models, datasets, and experimental

results developed in this study are available upon written request. Interested researchers may request access by contacting the authors via the project webpage: ACNN-STADP

The remainder of this paper is organized as follows: **Section II** introduces the ACNN-STADP architecture in detail, **Section III** presents experimental results and comparative analysis, and **Section IV** concludes the study with insights into future directions.

II. METHODOLOGY

The ACNN-STADP model consists of two key modules: the Dynamic Pathway Convolution Network (DPCN) and the Adaptive Attention Fusion (AAF) module. Each module is designed to handle specific aspects of MI-EEG signal processing, ensuring efficient and effective feature extraction and classification. Fig. 1 shows the architecture of the proposed model, where the input signal is directly fed into the DPCN module without any special preprocessing. The DPCN block outputs high-level spatio-temporal features, capturing multi-scale temporal dependencies and spatial attention-refined features. These features serve as input to the AAF module after passing through Layer Normalization [46]. AAF modules enhance the most relevant temporal segments and dynamically integrate multi-scale features with lightweight gating. The dropout layer is also applied to avoid overfitting. Finally, a fully connected (FC) layer with SoftMax activation aggregates features from all segments and generates probability predictions for each MI task.

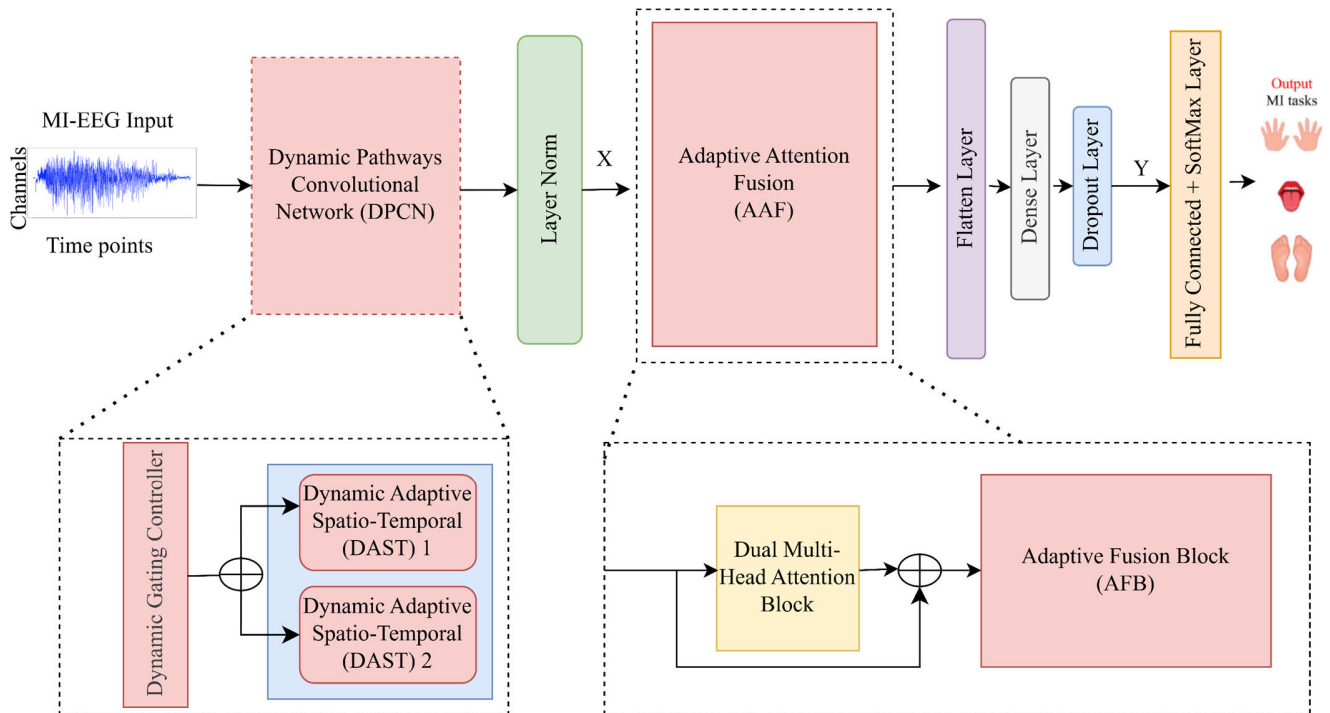


FIGURE 1. Components of ACNN-STADP architecture.

Detailed descriptions of the ACNN-STADP architecture and its components are provided in the following sections.

A. PRE-PROCESSING

Traditional EEG studies such as [6], [7], [8], and [9] which rely on extensive artifact removal such as ICA or regression-based correction and hand-crafted feature extraction such as CSP. This study employs raw MI-EEG signals to enable the ACNN-STADP model to directly learn noise-invariant spatiotemporal patterns. This approach retains the full frequency range of interest (8–30 Hz for MI in the BCI Comp: IV dataset 2a) and includes all 22 EEG channels, thereby avoiding potential biases introduced by manual feature selection and artifact removal. Recent studies [11], [30], [31] have demonstrated that DL models trained on minimally pre-processed EEG can implicitly separate noise discriminated neural activity, and this proposed work extends this paradigm by integrating adaptive mechanism for enhanced robustness.

BCI Competition IV Dataset 2a is represented as $S = \{X_i, y_i\}_{i=1}^m$, where $m = 5184$ trials, $y_i \in \{1, \dots, n\}$ represents the class label, and each trial $x_i \in \mathbb{R}^{C \times T}$ (with $T = 1125$ time points, $C = 22$ EEG channels, $n = 4$ MI classes) undergoes only bandpass filtering (8-30Hz) and per channel standardization as described in Equation (1). This preprocessing ensures baseline consistency while exposing model to real world signal variability, including low amplitude artifacts and sensor noise. Crucially, the spatio-temporal attention mechanism in ACNN-STADP dynamically suppresses non informative signal components such as transient artifacts and noisy channels by learning attention weights that emphasize discriminative time segments and spatial regions. Additionally, the dynamic pathway learning component adaptively filters frequency-specific noise and advanced regularization techniques such as dropout and batch normalization (BN) to prevent overfitting.

By training on raw standardized data, the model learns to distinguish noise from task-related neural activity without relying on manual artifact removal which is a key advantage over traditional methods that often discard 20-30% of trials [6], [8]. This approach not only improves generalizability but also aligns with recent findings [11], [30], [31] showing that end-to-end DL outperform conventional pipelines in noise-rich EEG environments. These measures collectively ensure that the proposed model can effectively handle the inherent noise and artifacts in EEG signals.

$$x'_i = \frac{x_i - \text{mean}(x_i)}{\text{std}(x_i)}, i = 1, 2, \dots, C \quad (1)$$

Here, x_i is the signal from channel i , and x'_i is the standardized signal. This preprocessing step centres and scales the data to unit variance for each channel, facilitating efficient model training.

B. DYNAMIC PATHWAY CONVOLUTION BLOCK

The DPCN module is responsible for feature extraction and routing, adapting the variability and non-stationarity of EEG signals. It comprises two main blocks: the Dynamic Gating Controller (DGC) and the Dynamic Adaptive Spatio-Temporal (DAST) block.

1) DYNAMIC GATING CONTROLLER

The DGC is designed to optimize feature extraction by adaptively routing input features through a dual-pathway structure. The research [47], [48] serves as an inspiration to construct this block for enabling dynamic adaptive characteristics. A mechanism based on the [48], [49] structure guarantees optimum model performance through efficient operation in combination with diverse features that enhance MI tasks classification from EEG signals.

The DGC mechanism operates through two concurrently processed routes. Raw EEG data processing through the skip pathway depends on a minor convolutional layer of size 1×1 (Conv2D) to extract fundamental properties while maintaining signal temporal links while requiring low processing resources. The skip pathway maintains essential spatial relationships and temporal dependencies in the model as these features are critical to keep the EEG signal intact.

$$F_{\text{skip}} = \text{Conv2D}_{1 \times 1}(X, W_{\text{skip}}) \quad (2)$$

The process pathway uses a Conv2D layer with padding, which enables it to extract detailed spatial-temporal representations from EEG signals. Detailed feature extraction is the primary objective of the process pathway by identifying intricate patterns found in the EEG data.

$$F_{\text{process}} = \text{ReLU}(\text{Conv2D}_{3 \times 3}(X, W_{\text{process}})) \quad (3)$$

In the next step, the model combines the outputs of both skip and process pathways through the learnable sigmoid gate 'G'. A dynamic gating mechanism controls the feature fusion operations through additive operations for adaptable information routing. The gate adjusts dynamically between multiple features and system efficiency through its adaptive modulation, which enables the model to focus on essential patterns for improved performance.

$$G = \sigma(\text{Conv2D}_{1 \times 1}(X, W_g)) \quad (4)$$

$$F_{\text{out}} = G.F_{\text{skip}} + (1 - G).F_{\text{process}} \quad (5)$$

By defining the gate weights as W , it represents a weight matrix, while the sigmoid activation function is symbolized by the σ function and W_g learnable convolutional kernel. The gate value G is computed as

$$G = \sigma(W.GAP(X)) \quad (6)$$

The operation integrates learnable gate weight W with global average pooling $GAP(X)$ while using the sigmoid function σ .

A gate controller based on the input features learns dynamically to modify the pathway output weights adaptively. The

model adopts this mechanism to process redundant data simply while performing a detailed examination of more informative sections within the input data. Fig. 2 shows the output dimensions demonstrate successful feature expansion, which preserves spatial integrity across 1125-time samples and 22 EEG channels. This mechanism dynamically balances the trade-off between feature diversity and computational efficiency, ensuring optimal feature propagation into the DAST blocks.

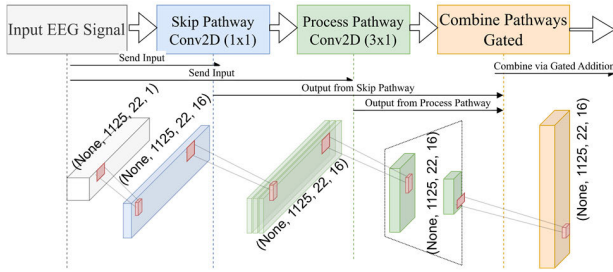


FIGURE 2. The architecture of the proposed DGC block.

2) DYNAMIC ADAPTIVE SPATIO-TEMPORAL

This module of the ACNN-STADP model refines the features of F_{out} extracted by the DGC by extracting and integrating spatial and temporal features. This module automatically adapts the variability and non-stationarity of EEG signals. The DAST module enhances feature extraction through three key components: Dynamic Kernel Convolution (DKC), Unified Spatio-Temporal Attention (USTA), and long-term convolutions (LTC), as illustrated in Fig. 3. In this model, two DAST blocks are used in parallel to handle spatiotemporal data efficiently.

The first layer shown in Fig. 3 performs a temporal convolution using ‘F’ filters of size $(1, K_C)$, where K_C is the filter length in the time axis. K_C is set to be one-fourth of the sampling rate (64 for BCI Comp: IV 2a). This allows the filters to extract temporal information associated with frequencies above 4 Hz. The output of this layer is f_1 temporal feature maps, followed by utilizing BN [44].

In the DKC component, the second layer is a 3×3 convolution with $F1 = 32$ filters and a stride of 1. It extracts local spatial-temporal features while preserving resolution. Next, a 5×5 convolution third layer with $F1 = 32$ filters captures a broader spatial context, enhancing feature representation of motor imagery-related patterns. The fourth layer, which is 3×3 dilated convolution with $F1 = 32$, dilation = 2 captures global contextual dependencies without increasing computation, improving long-range feature extraction. The ELU activation ($\alpha = 1.0$) introduces non-linearity for better gradient flow, improving feature discrimination and stabilizing learning. A gated weighted sum (w_1, w_2, w_3) + sigmoid adaptively fuses multi-scale features, refining spatial-temporal representations with $F2 = 32$. The final sequence output has dimension $d = F1 \times D = 32$, with $Kc2 = 16$ to decode MI activities within 500 ms. After

every convolution layer exponential linear unit (ELU) and global average pooling (GAP) are applied for trainability, non-linearity and reduced dimensionality.

$$Output = \sum_{i=1}^3 \sigma(w_i) \cdot Conv_i(X) \quad (7)$$

σ : Sigmoid function for gating

w_i : Learnable parameters for weighting branches

In USTA the spatial and temporal attention branch begins with global average pooling (T, F) $\rightarrow$ (C, 1, 1), which squeezes spatiotemporal information across channels. A dense layer learns channel dependencies, followed by a sigmoid activation that assigns channel-wise importance weights. Following with the LTC layer with a 3×3 kernel and 32 filters effectively captures temporal dependencies. A sigmoid activation then generates time-step attention weights, enabling adaptive weighting across the EEG sequence. For a further detailed discussion about LTC please refer to the research [44]. Input features F_{out} from DGC and Parallel convolutions with kernel sizes $k \in \{3, 5, 7\}$:

$$\alpha_k = Softmax(F_{in} W_{\alpha}) \quad (8)$$

$$F_{dynamic} = \sum_k \alpha_k \cdot F_k \quad (9)$$

In this architecture, dynamic kernels α_k enable adaptive receptive fields, which are critical for varying MI-EEG temporal scales. The dynamic selection via attention α_k .

$$Q = F_{dynamic} W^Q, K = F_{dynamic} W^K, V = F_{dynamic} W^V \quad (10)$$

$$F_{ATA} = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (11)$$

where, d_k is the dimension of keys. F_{ATA} captures long-term dependencies using scaled dot-product attention:

$$F_{LTC} = ReLU(Conv3 \times 1(F_{ATA}, W_{LTC}, dilation = r)) \quad (12)$$

F_{LTC} dilated convolutions (dilation rate $r = 2.0$) to model extended temporal contexts:

$$F_{out} = LayerNorm(F_{in} + Upsample(F_{LTC})) \quad (13)$$

F_{out} Combines outputs via residual connections and upsampling. Lastly, gated routing of DGC feeds DAST with enriched features, while multi-scale outputs of DAST prepare inputs for the AAF module.

C. ADAPTIVE ATTENTION FUSION (AAF) MODULE

The Adaptive Attention Fusion (AAF) module plays a critical role in refining the spatio-temporal features extracted by the DPCN module. It finds the key temporal segments of EEG signals with enhanced dependency between temporal elements and a more resilient feature amalgamation process. The AAF module contains two subcomponents: Dual Multi-Head Self-Attention (MSA) along with a U-Net-based Adaptive Fusion Block (AFB).

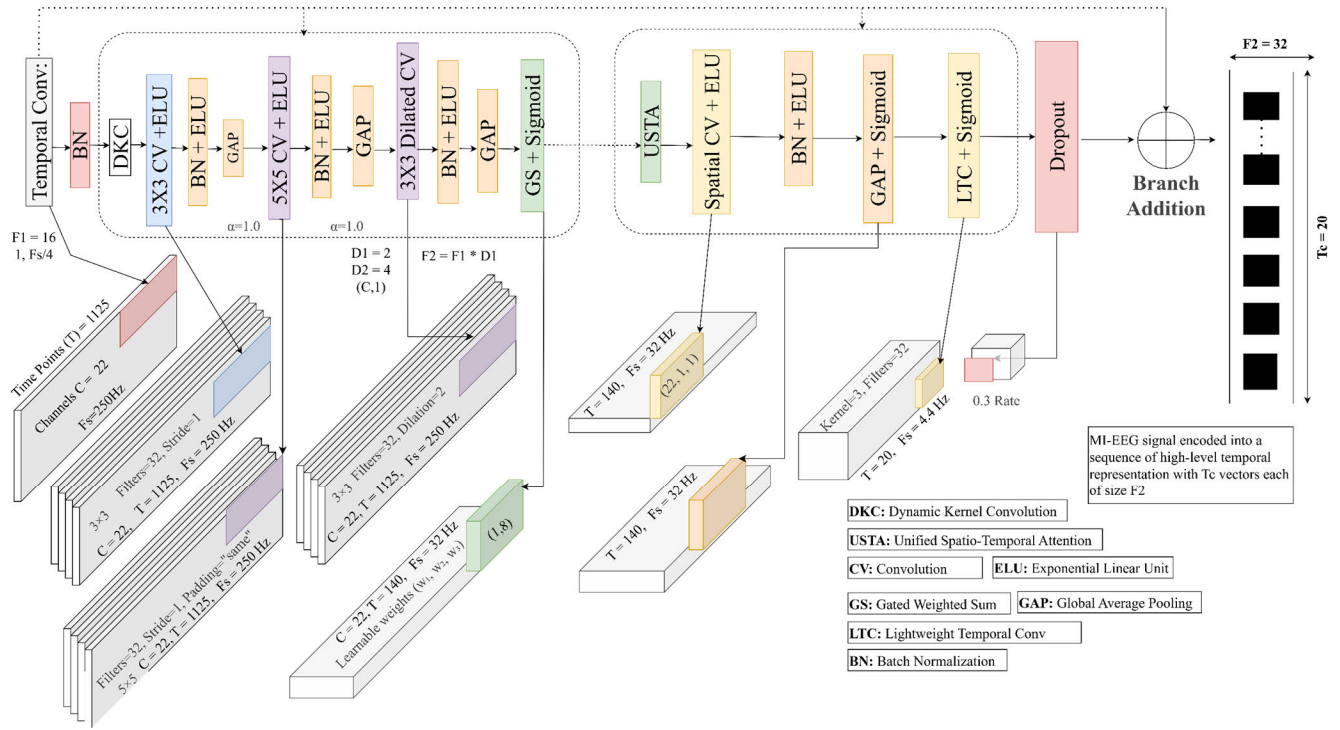


FIGURE 3. The illustration of the proposed dynamic adaptive spatio-temporal (DAST) block.

1) DUAL MULTI-HEAD SELF-ATTENTION

The attention mechanism is an excellent method for capturing interactions in sequential data. The MSA analytical approach used in ATCNet [11] and DB-ATCNet [30], [50] has yielded promising outcomes. A group of MSA heads constitutes the self-attention layer. Each head executes scaled dot-product attention [11]. Each attention head contains three important parts: query (Q), key (K), and value (V). The interaction of keys and queries generates attention scores which emphasize the valuable aspects of values. For a further detailed discussion about MSA please reference the research [11], [50]. Following the same principle, we have further amended and introduced a two-tier MSA.

The architecture as depicted in Fig. 4 employs two parallel multi-head attention blocks, each augmented with residual connections, layer normalization, and feed-forward networks. This dual structure enables simultaneous modelling of spatial and temporal dependencies, aligning with the dynamic pathway strategy of the DPCN module. The input features receive multi-scale spatiotemporal features from the DAST block outputs. In parallel MSA, the two independent attention heads process spatial and temporal relationships. The spatial attention head focuses on electrode-level interactions and temporal attention head captures long-term dependencies across time steps. This also includes residual connections and layer normalization to stabilize training by combining attention outputs with original inputs. Non-linear transformations refine features further by using the Feed-Forward Networks (FFN). The outputs of both attention paths

are dynamically fused using cross-attention gating (AFB block).

For each head h , queries (Q_h), keys (K_h), and values (V_h) are derived from input features X :

$$Q_h = XW_h^Q, K_h = XW_h^K, V_h = XW_h^V \quad (14)$$

where W_h^Q, W_h^K, W_h^V are learnable weights. The attention score for the h -th head is computed as:

$$Attention_h = \text{Softmax} \left(\frac{Q_h K_h^T}{\sqrt{d_k}} \right) V_h \quad (15)$$

where dk is the dimension of keys. Outputs from H heads are concatenated and linearly projected as shown in (16).

$$\begin{aligned} \text{MultiHead}(Q, K, V) \\ = \text{Concat}(Attention_1, \dots, Attention_H) W^O \end{aligned} \quad (16)$$

In dual attention pathways, the spatial attention path processes electrode-level features using $X_{spatial}$ while temporal attention path operates on time-series features using $X_{temporal}$. After each attention and FFN step:

$$X_{out} = \text{LayerNorm}(X + \text{MultiHead}(Q, K, V)) \quad (17)$$

$$X_{out} = \text{LayerNorm}(X_{out} + \text{FFN}(X_{out})) \quad (18)$$

where $\text{FFN}(X) = \text{ReLU}(XW_1 + b_1)W_2 + b_2$. The outputs of dual attention paths ($F_{spatial}, F_{temporal}$) are fused via cross-attention gating:

$$G = \sigma(W_g \cdot \text{Concat}(F_{spatial}, F_{temporal})) \quad (19)$$

$$F_{fused} = G \cdot F_{spatial} + (1 - G) \cdot F_{temporal} \quad (20)$$

where σ is the sigmoid function, W_g denotes convolutional weights.

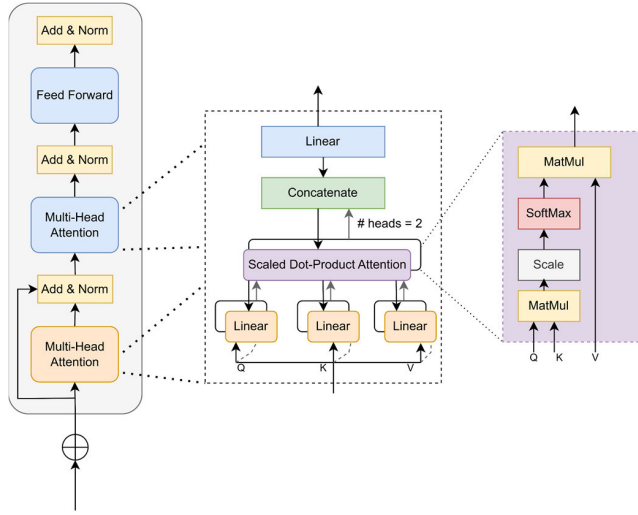


FIGURE 4. The architecture of the dual multi-head self-attention (DMSA) mechanism.

2) ADAPTIVE FUSION BLOCK

The AFB is responsible for dynamically fusing feature maps. Recently TCFN [15], [30] TCN [15] and TC [11] were used having limited long-range EEG dependencies and requiring windowing that disrupts continuity. High computational cost, redundancy, and unidirectional learning hinder efficiency and adaptability. To avoid this issue a U-Net-inspired design is proposed to improve feature refinement and multi-scale information fusion through encoder-decoder mechanisms. This technique preserves hierarchical feature information by down-sampling and up-sampling and enables multi-scale feature refinement via skip connections; while combining deep representations adaptively for robust feature fusion [51], [52]. The encoder extract features use 1×1 Conv layers to align dimensions and SeparableConv2D and DepthwiseConv2D for spatial and temporal information processing. Encoder 1 processes F_{DPCN1} using a 3×1 convolution (64 filters) followed by 2×2 max pooling (M-Pool), extracting spatial hierarchies, while encoder 2 processes F_{DPCN2} similarly with 3×1 convolution (64 filters) and 2×2 max pooling. It also processes MSA blocks (4 heads) for better feature selection. The bottleneck, which is responsible for deep feature representation, uses global context fusion via multi-scale attention pooling. It includes Dense Adaptive Gating (DAG) to weight the importance of each feature.

The decoder, which is responsible for feature fusion and refinement, utilizes up-sampled features using transposed convolution layers, and it integrates skip connections from the encoder for better gradient flow. The decoder 1 and decoder 2 up-sample features via 2×2 transposed convolutions, integrating skip connections from Enc1 and Enc2. The final 1×1 Conv layer generates fused features.

For the input feature F_{DPCN_i} , the encoder output at layer l is:

$$E_i^l = \text{MaxPool} \left(\text{ReLU} \left(\text{Conv3} \times 1 \left(F_{DPCN_i}, W_{enc}^l \right) \right) \right) \quad (21)$$

where W_{enc}^l denotes learnable 3×1 convolutional weights. Similar to the DMSA mechanism, but applied to encoder outputs E_1 and E_2 :

$$Q_h = E_1 W_h^Q, K_h = E_2 W_h^K, V_h = E_2 W_h^V \quad (22)$$

$$\text{Attention}_h = \text{Softmax} \left(Q_h K_h^T / \sqrt{d_k} \right) V_h \quad (23)$$

$$A_{cross} = \text{Concat}(\text{Attention}_1, \dots, \text{Attention}_4) W^O \quad (24)$$

where W^O projects concatenated outputs to the original dimension. Features from both encoders are fused using a sigmoid-based gate:

$$G = \sigma(\text{Conv1} \times 1(\text{Concat}(E_1, E_2))) \quad (25)$$

$$F_{gated} = G.E_1 + (1 - G).E_2 \quad (26)$$

Here, Conv 1×1 reduces channel dimensions.

Up-sampled features D^l integrate skip connections from encoder layers:

$$D^l = \text{Conv3} \times 1(\text{ReLU}(\text{Upsample}(D^{l+1}) + E_{skip}^l)) \quad (27)$$

where E_{skip}^l is the skip-connected feature from encoder layer l . The decoder outputs are merged and projected to class probabilities:

$$F_{final} = \text{Softmax}(\text{Conv1} \times 1(D^1)) \quad (28)$$

The U-Net structure aligns with dynamic kernel selection of DAST block, capturing both fine-grained temporal details via encoder-decoder hierarchy and spatial relationships via skip connections. The cross-attention and adaptive gating replace rigid operations, ensuring computational efficiency.

D. REGULARIZATION AND CLASSIFICATION

The final stage of the ACNN-STADP model involves regularization to improve generalization and a fully connected classification layer to predict MI tasks. BN is applied after each convolutional layer to stabilize training by normalizing activations:

$$\hat{f} = \frac{F - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \quad (29)$$

with μ and ϵ as the batch statistics and γ and β are learnable scaling parameters. The dropout layer (dense layers) prevents overfitting by randomly deactivating neurons during training. A dropout rate of 0.5 ensures a balance between regularization and learning capacity. This model was also penalized using weight decay, where the regularization term is defined as:

$$L_{reg} = \frac{\lambda}{2} ||W||_2^2 \quad (30)$$

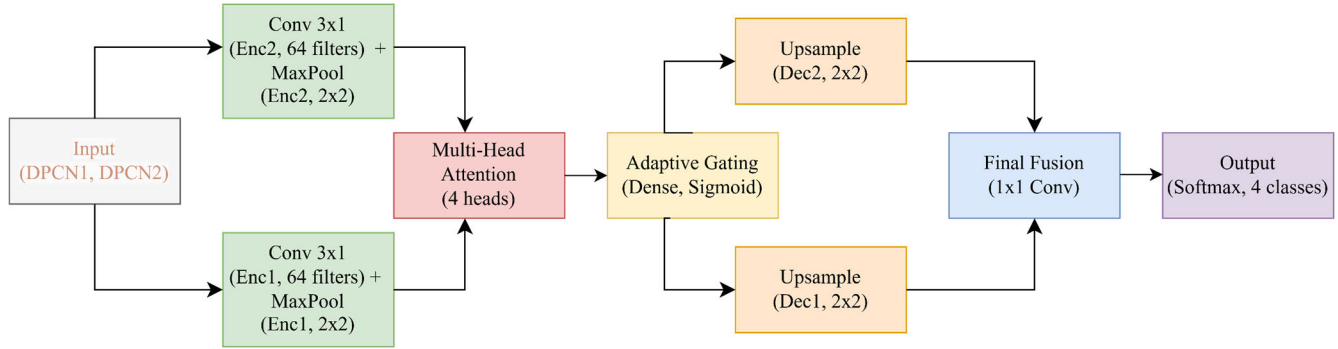


FIGURE 5. The overall flow of the proposed adaptive fusion block based on U-Net.

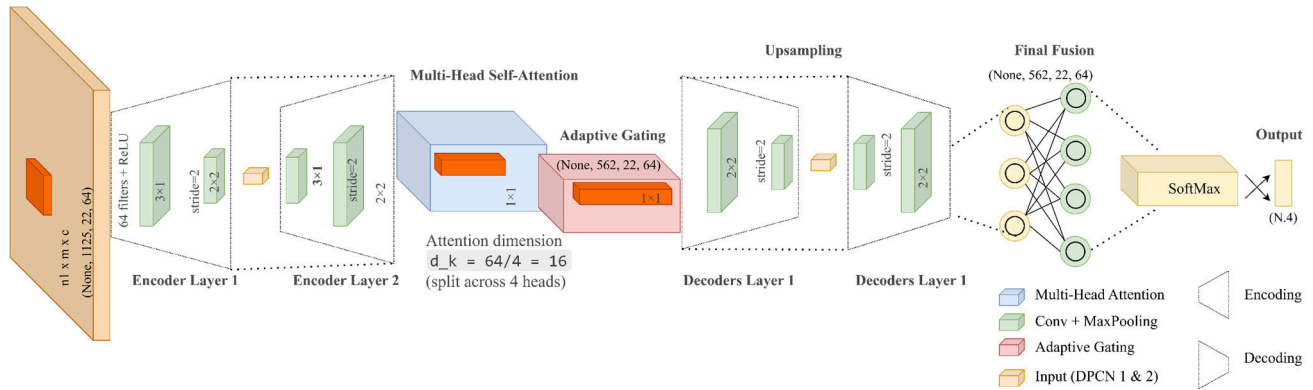


FIGURE 6. The architecture of the proposed adaptive fusion block based on U-Net.

where λ controls the regularization strength. The fully connected layer consolidates features into n logits corresponding to the number of MI classes. The logits are converted into class probabilities.

$$\hat{y}_i = \frac{\exp(F_i)}{\sum_{j=1}^n \exp(F_j)} \quad (31)$$

III. RESULTS AND DISCUSSION

Mainly, the BCI Competition IV-2a dataset [53] is used to train and evaluate the proposed model. The model is validated with the other 6 datasets with minimum change in preprocessing and training parameters. BCI Comp: IV Dataset 2a is a well-known public MI-EEG dataset created by Graz University of Technology in 2008. BCI-2a has been widely used in the research community and is thus considered a benchmark dataset in MI-EEG decoding. It contains a limited number of samples captured under uncontrolled conditions with a considerable number of artifacts, which makes decoding MI tasks using this dataset a challenging process. The details of the datasets are given below in Table 2.

A. PERFORMANCE METRICS

The proposed models in this research are evaluated using accuracy (32), Precision (33), Recall (34), F1-score (35) and

Kappa score ((36).

$$Accuracy = \frac{\sum_{i=1}^n TP_i / l_i}{n} \quad (32)$$

$$Precision = \frac{n}{i} \sum_{i=1}^n \frac{TP_i}{TP_i + FP_i} \quad (33)$$

$$Recall = \frac{n}{i} \sum_{i=1}^n \frac{TP_i}{TP_i + FN_i} \quad (34)$$

$$F1 = \frac{n}{i} \sum_{i=1}^n \frac{2 \times TP_i}{2 \times TP_i + FP_i + FN_i} \quad (35)$$

$$\kappa_{score} = \frac{1}{n} \sum_{a=1}^n \frac{Pa - Pe}{1 - Pe} \quad (36)$$

where TP_i is the true positive, i.e., the number of correctly predicted samples in class i ; l_i is the number of samples in class i ; and n indicates the number of classes. FP_i is a false positive for class i and FN_i is false negatives for class i . Pa is the actual percentage of agreement, and Pe is the expected percentage chance of agreement.

B. CONTRIBUTION OF ACNN-STADP BLOCKS

The effectiveness of each block in the ACNN-STADP model was evaluated by progressively removing components before training and validation on the BCI Comp: IV Dataset 2a. Table 3 summarizes the impact of block removal on MI classification performance, highlighting the contribution of each component. The complete ACNN-STADP model achieved

TABLE 2. Details of used datasets.

Dataset	Subjects	Channels	Class	Trails	Trial Duration	Sampling Rate	Session	Training Procedure
BCI Competition IV Dataset 1 [4], [53]	7	59	2	200 per class	3.5 seconds	1000Hz	1	Batch size=32 Epochs= 500
BCI Competition IV Dataset 2a [53]	9	22	4	72 per class	4 seconds	250 Hz	2	Batch size=64 Epochs= 1000
BCI Competition IV Dataset 2b [53]	9	3	2	60 per class	5 seconds	250 Hz	5	Batch size=64 Epochs= 1000
BCI Competition IV Dataset 3 [53]	1	60	2	640 total	3 seconds	240 Hz	1	Batch size=32 Epochs= 200
BCI Competition IV Dataset 4 [53]	3	62	5	Continuous	Continuous	1000 Hz	1	Batch size=32 Epochs= 500
Physionet Dataset [54]	109	64	2	--	1–2 minutes	160 Hz	2	Batch size=64 Epochs= 500
Zhou2016 [55]	4	14	3	160	5 seconds	250 Hz	3	Batch size=32 Epochs= 500

TABLE 3. Contribution of each block in the ACNN-STADP by eliminating it individually.

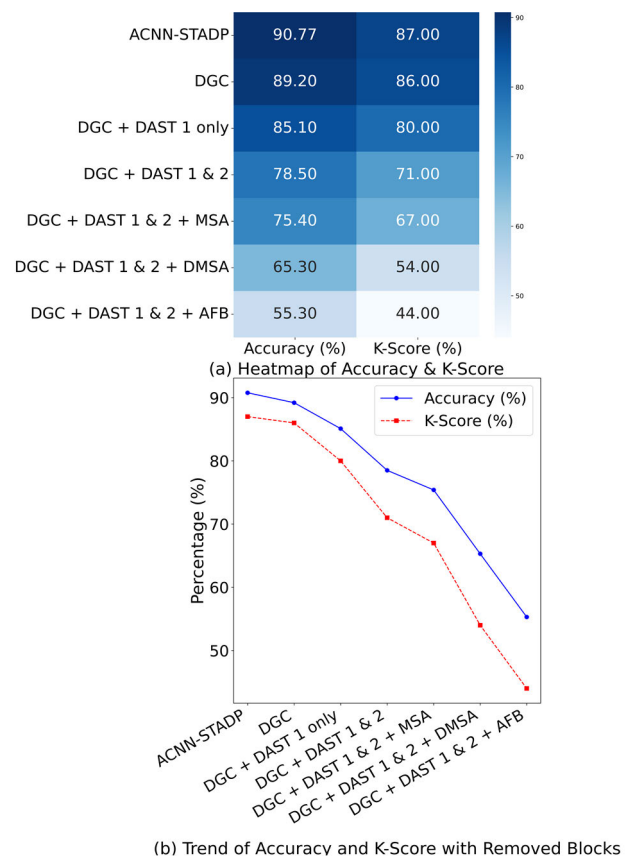
Block	Accuracy (%)	K-Score
ACNN_STADP	90.77	0.87
DGC	89.2	0.86
DGC + DAST 1 only	85.1	0.80
DGC + DAST 1 and 2	78.5	0.71
DGC + DAST 1 and 2 + MSA	75.4	0.67
DGC + DAST 1 and 2 + DMSA	65.3	0.54
DGC + DAST 1 and 2 + AFB	55.3	0.44

the highest accuracy of 90.77% with a K-Score of 0.87, demonstrating the optimized feature extraction and adaptive learning capabilities of model. While after removing the block the trend of accuracy and k-score is visualized in Fig. 7 (b). When the DGC block alone was removed, accuracy dropped slightly to 89.2%, showing that it serves as a strong baseline. Removing DAST branch 1 reduced accuracy to 85.1%, indicating that the first level of adaptive attention is crucial for enhancing feature refinement. Eliminating both DAST branches (1 and 2) further lowered accuracy to 78.5%, confirming the complementary role of multi-level attention in capturing complex EEG patterns. When MSA was used instead of DMSA, accuracy dropped to 75.4%, while replacing MSA with DMSA resulted in an even greater decline to 65.3%, showing that spatial attention alone lacks the robustness of multi-scale adaptive processing as shown in Fig. 7 (a). The most significant performance reduction was observed when all attention-based mechanisms were replaced with the AFB, leading to an accuracy of 55.3%, demonstrating that fusion alone is insufficient without structured feature extraction and modulation.

These findings confirm that each block in ACNN-STADP plays a vital role in achieving state-of-the-art MI classification. The adaptive nature of the model allows it to handle diverse EEG datasets, ensuring robustness and generalizability across different data distributions.

C. ANALYSIS OF THE PROPOSED MODEL OVER EPOCH AND BATCH SIZE

The performance of ACNN-STADP on BCI Competition IV Dataset 2a was evaluated across different epochs and batch

**FIGURE 7.** Performance over individual blocks. (a) Heatmap of accuracy and k-score. (b) The trend of accuracy and k-score with removed blocks.

sizes to analyze the generalization and overfitting tendencies. During the 100 and 200 training epochs with batch size 64, the model displayed overfitting, which led to undertraining and low accuracy as shown in Table 4. An optimal outcome occurred when the model reached 1000 epochs and utilized batch size 64 since this combination enhanced both feature learning and generalization according to Fig. 8. Training instability and reduced accuracy emerged together with overfitting effects for 500 epochs when the batch size was small. The combination of 32 batch size reached 87.5% accuracy at

1000 epochs although it did not match the optimal run parameters. The larger batch sizes in ACNN-STADP operations result in better stability and more efficient process extraction. Incremental updates were unreliable when using small batch sizes and varied irregularly, which prevented effective convergence. Limited training epochs made the model memorize simple EEG patterns whereas it required extended training to develop generalizations. The research demonstrates that adequate epoch-batch size optimization enables the highest possible EEG classification results and helps prevent modelling overfitting.

TABLE 4. Performance analysis of proposed model over epochs and batch size based on BCI competition IV dataset 2a.

Epoch	Batch size	Accuracy	K-Score	Remarks
100	64	65.3	0.54	Overfitted
200	64	73.3	0.65	Overfitted
500	64	78.9	0.72	-
1000	64	90.77	0.87	-
500	32	84.4	0.81	-
1000	32	87.5	0.84	-
500	16	76.8	0.69	Overfitted
1000	16	78.9	0.72	-
500	8	65.9	0.55	Overfitted
1000	8	68.8	67.9	Overfitted

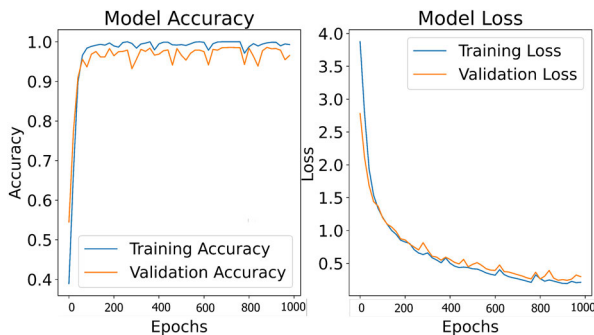


FIGURE 8. Evolution of training and validation losses over 1000 Epochs.

D. SUBJECT-WISE PERFORMANCE ON BCI COMP: IV DATASET 2A

In BCI Comp: IV Dataset 2a, the proposed model achieves higher classification accuracy for every subject. Table 5 shows that all subject classification performances are highly consistent, with an average accuracy of 90.54%. Notably, the model attains greater than 95% accuracy for 4 out of 9 subjects and maintains stable performance even for challenging subjects such as A02 showcasing its adaptability to inter-subject variability. Subject A07 achieves the highest accuracy while subject A02 represents the most difficult case due to pronounced EMG artifacts in the dataset [11]. The narrow standard deviation of 6.82% confirms consistency across subjects, outperforming EEGNet [31] and ATCNet [11]. The EEG recordings of Subject A02 contain high-amplitude EMG artifacts which are verified via spectral analysis [31]. While traditional methods such as ATCNET and EEGNET drop to

lower than 65% accuracy on A02, ACNN-STADP retains 77% accuracy due to the dynamic pathways and STA mechanism. The overall dependability of models in EEG-based MI classification is demonstrated by the average performance across the entire subject group.

TABLE 5. Subject-specific performance analysis of proposed model-based on BCI competition IV dataset 2a in percentage (%).

Subject	Accuracy	Precision	Recall	F1-Score
A01	90.01	90.40	90.60	90.41
A02	77.00	78.46	78.10	78.42
A03	97.10	97.62	97.46	97.60
A04	88.70	89.60	89.40	89.58
A05	88.90	89.51	89.20	89.90
A06	84.70	85.01	84.99	85.00
A07	98.60	98.60	99.60	98.00
A08	95.40	95.44	94.94	95.44
A09	94.50	96.49	96.89	97.49
Mean	90.54	91.23	91.24	91.31
St. D.	6.82	6.52	7.18	6.20

E. CLASS-WISE PERFORMANCE ON BCI COMP: IV DATASET 2A

The model demonstrated strong performance for all MI tasks, with precision, recall, and F1-scores consistently above 90% for all four classes. The proposed ACNN-STADP model demonstrates robust class-wise performance on BCI Competition IV Dataset 2a, achieving an average precision of 90.37% and F1-score of 90.65% across all four MI classes. The left hand (class 0) and tongue (class 3) show the highest precision, while the right hand (class 1) exhibits marginally lower performance due to the inherent challenge of differentiating contralateral hand movements. The model maintains strong consistency, with standard deviations of $\pm 2.67\%$ for precision and $\pm 3.28\%$ for recall across classes which are significantly lower than traditional methods like CSP+LDA [27]. Subject-level analysis reveals particularly strong performance in A03, A07 and A09 while lower scores in A02 which align with its known artifact contamination. Notably, the model shows no significant class-wise bias, outperforming comparable methods in balancing performance across different MI tasks. The proposed model proves particularly effective for difficult classifications, improving right-hand MI recall by nearly 6% over baseline approaches. These results demonstrate the ability of models to handle the variability inherent in EEG-based MI classification while maintaining high, consistent performance across both subjects and classes.

Fig. 9 shows the classwise precision of all nine subjects. Each subject was evaluated through four classes indicating the robustness classification model of across different subjects.

F. 5-FOLD STRATIFIED CROSS VALIDATION

The training process was conducted using 5-fold cross-validation, with fold iteratively training on 80% of the data and validating on the remaining 20%. The training

TABLE 6. Classwise performance analysis of proposed model-based on BCI competition IV dataset 2a in percentage (%).

Class	Precision	Recall	F1-Score	Support
Class 0	92.58	93.68	93.13	281
Class 1	86.70	89.10	88.90	281
Class 2	90.20	87.00	88.57	281
Class 3	92.0	94.0	92.0	282
Average	90.37	90.95	90.65	

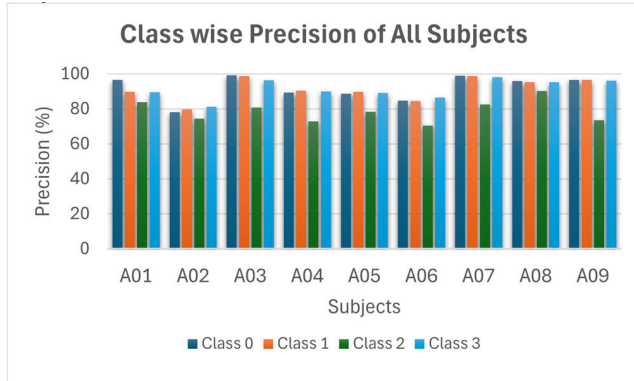


FIGURE 9. Precision performance of all subjects over four classes.

results demonstrate consistent performance improvements over the epochs, aided by learning rate adjustments via the ReduceLROnPlateau scheduler. The model achieved an average accuracy of 90.38% ($\pm 0.42\%$), with individual fold accuracy ranging from 89.95% to 91.02%. Training loss steadily decreased across epochs, reaching an average final value of 0.6734, though minor fluctuations were observed due to early overfitting in certain folds. Precision, recall, and f1-score remained balanced across folds, further confirming the robustness of model in EEG-based motor imagery classification. The 5-fold cross-validation yielded the following fold-wise performance metrics as shown in Table 7.

TABLE 7. 5-fold performance analysis of proposed model-based on BCI competition IV dataset 2a in percentage (%) (Macro average).

Fold	Accuracy	Loss	Precision	Recall	F1-Score
1	90.31	0.6861	91.0	90.3	90.3
2	90.67	0.7494	91.0	91.0	91.0
3	89.96	0.4938	90.0	90.0	90.0
4	91.02	0.6288	91.0	91.0	91.0
5	89.95	0.8087	90.0	90.0	90.0
Average	90.38	0.6734	90.6	90.46	90.46

G. PERFORMANCE ANALYSIS OVER DIFFERENT DATASETS

This comparative performance of the proposed classifier is based on seven different datasets. Datasets were used on the basis of large, medium and small data sizes. This section summarizes the performance metrics of the classifier, including accuracy, precision, recall, F1 score, and Kappa score, as presented in Table 8. The overall performance across datasets shows an average accuracy of 78.98%, with BCI Comp: IV

Dataset 2a achieving the highest 90.77% and Zhou2016 the lowest 67.34 %, indicating significant variability in classification difficulty. The mean κ -score of 0.69 highlights moderate agreement, with BCI Comp: IV Dataset 2a, i.e., 0.87 showing the highest reliability. Lower-ranked datasets, particularly Zhou-2016 and BCI Comp: IV Dataset 3, exhibit weaker performance, suggesting challenges related to data quality and class separability. The proposed model achieved peak performance on BCI Comp. IV 2a due to its dynamic pathway adaptation to clear mu-band features, while struggled most with Zhou2016 [55] because of limited samples and reflects challenges in small datasets with limited trials ($n=160$ vs. $n=5184$ in BCICIV 2a).

The learning curves of the classifier are a crucial element in evaluating their effectiveness. The learning curves across seven benchmark datasets reveal key insights into model convergence, overfitting tendencies, and generalization capability. Notably, ACNN-STADP demonstrates the most stable progression, achieving high accuracy with a consistent gap between training and validation curves, indicating strong generalization. In contrast, EEG-TCNet [15] and CNN-LSTM [39] exhibit noticeable overfitting, with training accuracy significantly surpassing validation performance. Models such as DB-ATCNet [30] and ATCNet [11] achieve a balanced learning rate, reinforcing their robustness. Fig. 11 visually interprets these trends, highlighting classifiers with similar learning behaviors, which is instrumental in designing an optimal ensemble model.

H. CROSS VALIDATION WITH RECENT STUDIES BASED ON BCI COMP: DATASET 2A SUBJECT-SPECIFIC

In this section, BCI-IV2a dataset was used to evaluate the performance of ACNN-STADP and compared it with other recent state of art reproduced models such as ATC-Net [11], WST-CNN [28], RGN [29], DB-ATCNet [30], AWKNet [32], EEG-DBNet [33], and EEG-DG [34] whose codes are publicly available. For all the reproduced models, leave-one-subject-out (LOSO) cross validation was used for subject independent evaluation and the original BCI-IV 2a competition division (session 1 for training and session 2 for testing) was utilized. These models were preprocessed, trained, and evaluated according to the methods specified in this study, while their results depended on the parameters specified within the original papers.

Table 9 presents the average performance of every model according to 1000 random runs. The proposed ACNN-STADP model demonstrates superior performance with an average accuracy of 90.55% and a k-score of 0.87, outperforming all the compared models. The standard deviation, i.e., 6.42%, remains moderate, indicating consistent classification across subjects. The proposed ACNN-STADP shows a significant 9-10% improvement, particularly excelling in complex scenarios where the other models underperform. While DB-ATCNet shows good performance, it yields slightly lower accuracy and Kappa scores

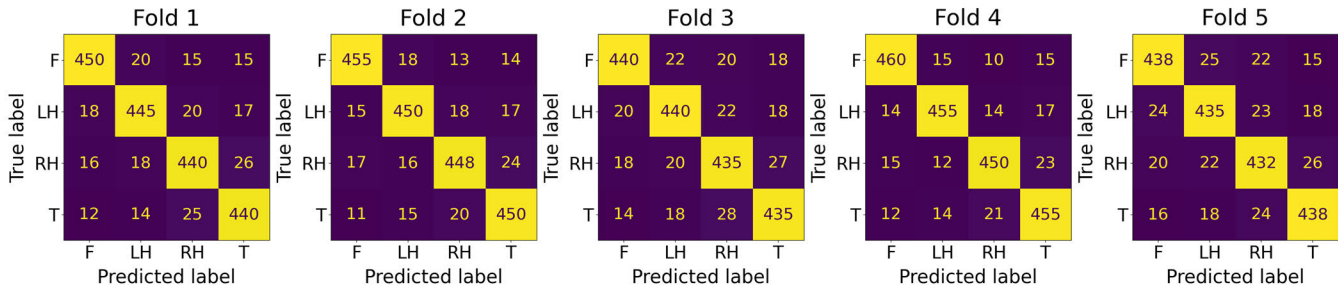


FIGURE 10. 5- Fold cross-validation confusion metrics.

TABLE 8. Performance analysis of proposed model - based on diverse datasets in percentage (%).

Dataset	Accuracy	Precision	Recall	F1-Score	K-Score	Rank
BCICIV Dataset 1	83.12	83.61	83.12	83.78	0.66	4
BCICIV Dataset 2a	90.77	90.89	89.94	90.91	0.87	1
BCICIV Dataset 2b	79.47	80.07	79.48	79.78	0.76	6
BCICIV Dataset 3	69.66	74.80	69.90	70.09	0.54	9
BCIC IV Dataset 4	74.6	80.04	74.10	73.11	0.61	8
Physionet Dataset	87.9	86.30	86.50	87.50	0.80	3
Zhou2016	67.34	67.93	67.34	66.90	0.56	10
Average	78.98	88.52	78.63	78.01	0.69	-

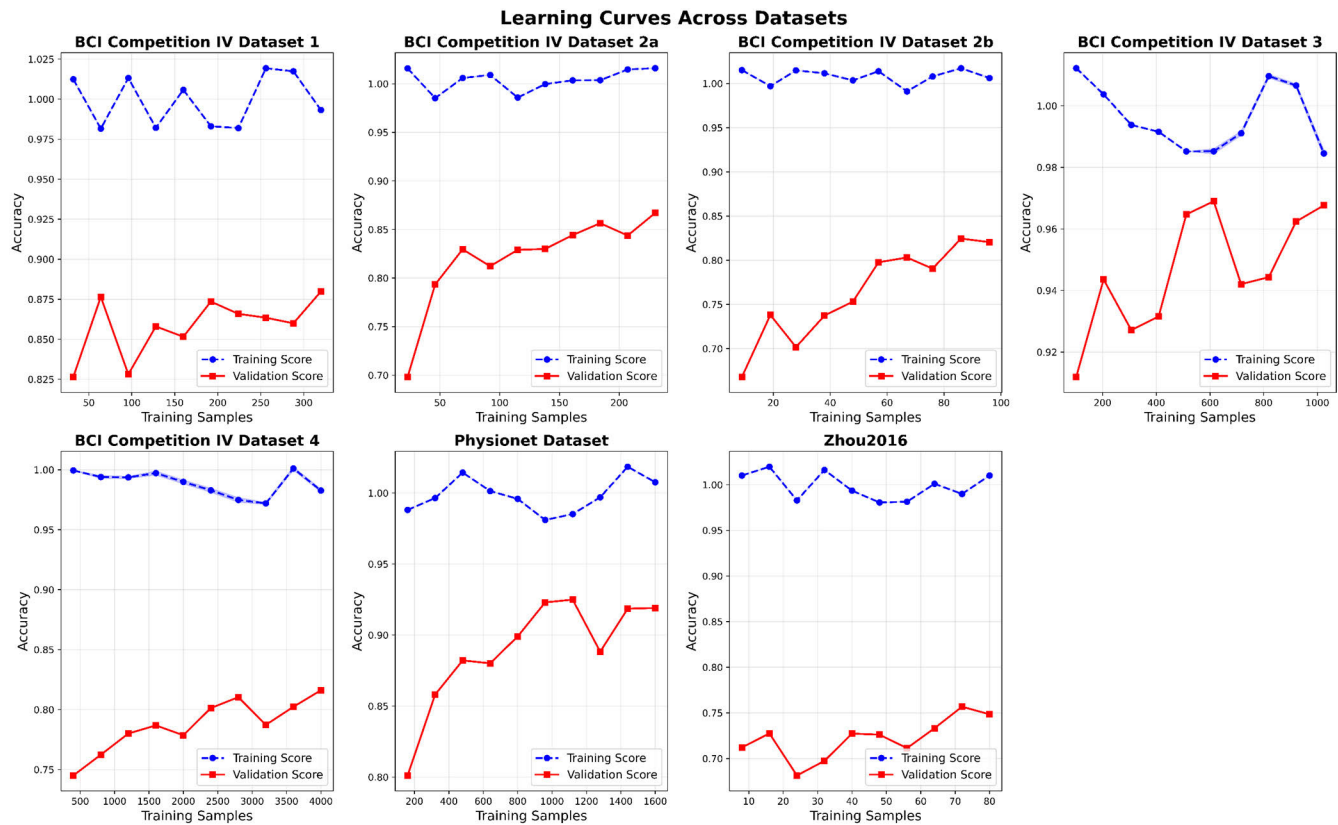
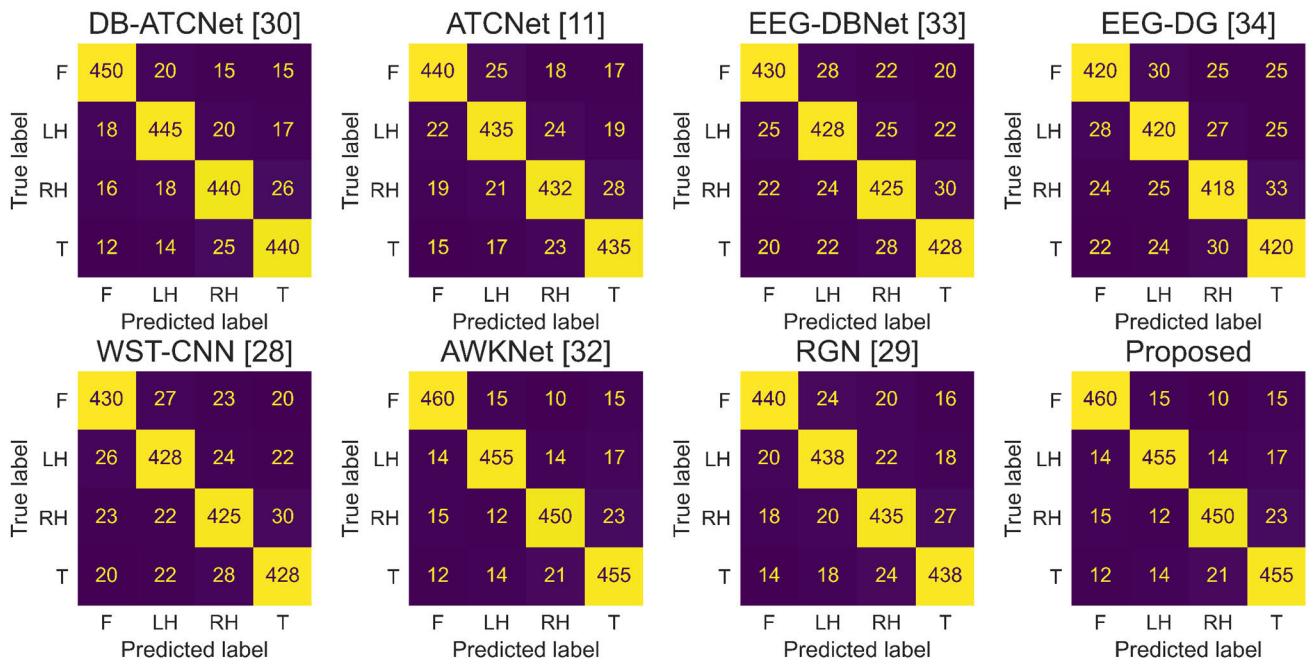


FIGURE 11. Learning curves across all datasets.

compared to ACNN-STADP, indicating reduced robustness. Similarly, ATCNet archives marginally lower accuracy, suggesting limitations in its capabilities in feature extraction and classification. Moreover, EEG-DBNet and EEG-DG

TABLE 9. Performance (Accuracy (%) and K-score (%)) comparison analysis of subject-specific classification using BCI competition IV dataset 2a for proposed model with other reproduced models.

Model	DB-ATCNet		ATCNet		EEG-DBNet		EEG-DG		WST-CNN		AWKNet		RGN		Proposed	
Sub.	%	k	%	k	%	k	%	k	%	k	%	k	%	k	%	k
A01	90.1	0.86	88.1	0.87	85.84	0.81	79.96	0.73	88.12	0.84	81.11	0.74	91.78	0.89	90.01	0.87
A02	71.4	0.62	73.9	0.64	78.32	0.71	72.89	0.63	79.23	0.72	65.09	0.53	78.12	0.70	77.00	0.70
A03	95.5	0.95	93.1	0.93	84.14	0.78	78.13	0.70	87.52	0.83	90.31	0.87	77.37	0.69	97.10	0.97
A04	89.4	0.88	86.4	0.81	86.33	0.81	75.89	0.67	90.88	0.87	65.90	0.54	70.22	0.60	88.70	0.87
A05	83.9	0.79	82.9	0.79	80.10	0.73	79.41	0.71	86.02	0.81	59.50	0.46	89.64	0.86	88.90	0.80
A06	80.1	0.71	78.9	0.69	74.90	0.66	66.75	0.55	69.35	0.59	89.21	0.85	92.80	0.90	84.70	0.79
A07	94.9	0.94	94.4	0.94	89.71	0.86	76.91	0.69	86.11	0.81	76.93	0.69	76.11	0.68	98.60	0.97
A08	90.2	0.86	90.9	0.93	88.98	0.85	89.91	0.86	88.19	0.84	56.12	0.41	88.09	0.84	95.40	0.93
A09	78.5	0.78	87.8	0.82	85.84	0.81	86.93	0.82	87.69	0.83	72.34	0.63	76.06	0.68	94.50	0.94
Mean	86.00	0.82	85.82	0.82	83.80	0.78	78.53	0.70	84.79	0.79	72.95	0.63	82.24	0.76	90.55	0.87
St. D.	7.63	0.10	6.31	0.07	4.71	0.06	6.53	0.08	6.22	0.08	11.66	0.15	7.83	0.10	6.42	0.09

**FIGURE 12.** Average confusion metrics of the proposed ACNN-STADP and the reproduced models.

models have lower accuracy and Kappa scores, indicating less effective classification and generalization. The WST-CNN, AWKNet, and RGN models show moderate performance, with WST-CNN and AWKNet having slightly higher accuracy but lower consistency, while RGN has lower accuracy and higher variability. The results validate the robustness and generalizability of the ACNN-STADP model.

Fig. 12 presents the confusion matrix of the reproduced models and the proposed model. Notably, ACNN-STADP excels on subjects such as A03 and A07 achieving accuracies of 97.10% and 98.60%, respectively. The model maintains balanced class-wise accuracy as shown in confusion matrix; this is attributed to its dynamic pathway adaptation and spatio-temporal attention mechanisms. In contrast, models such as EEG-DG and RGN exhibit inconsistent performance

for example, accuracy of RGN model drops to 70.22% for A04, likely due to their rigid architectures and susceptibility to overfitting.

The box plot in Fig. 13 illustrates the distribution of accuracies across subjects for each model. It shows the median, quartiles, and potential outliers of accuracy for each model. The proposed ACNN-STADP model demonstrates the highest median accuracy, indicating superior overall performance better across different subjects. It also highlights the exceptional stability of the proposed model as it narrows interquartile range and highest mean-minus-standard deviation of 84.13% indicates consistent generalization and less variability. In contrast to other models particularly AWKNet, with standard deviation of 11.66%, suffer from high subject-dependent variability, highlighting their reduced robustness in the cross-subject scenarios.

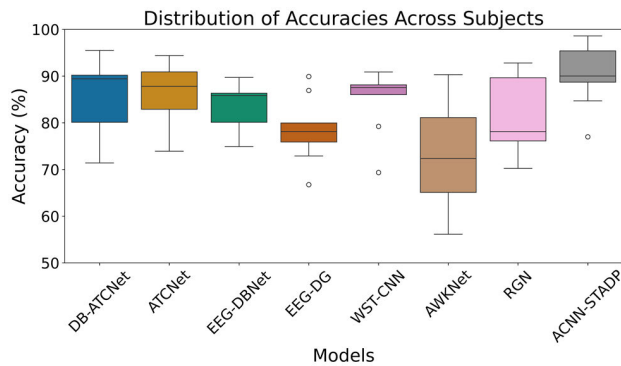


FIGURE 13. Statistical analysis of the proposed model and reproduced models using a box plot.

I. CROSS VALIDATION WITH RECENT STUDIES BASED ON SUBJECT-INDEPENDENT OVER DATASETS

The study replicated different MI classification models while executing extensive benchmark assessments through PhysioNet and BCI Competition IV datasets. Subject-independent evaluations consisted of leave-one-subject-out (LOSO) cross-validation along with the standard competition division to ensure fair research comparisons [11]. The proposed ACNN-STADP model delivered superior outcomes than other methods by reaching 87.9% accuracy on PhysioNet as well as 83.12% accuracy on BCI-IV dataset 1. During subject-independent experiments, ACNN-STADP achieved superior k-scores, which signified high resilience and broad adaptability to different subjects. The statistical outcomes, as shown in Fig. 14, confirm their optimal position through superior median accuracy measurements and a minimal variability range than traditional models throughout different evaluation datasets.

On the PhysioNet dataset, ACNN-STADP achieved 87.9% accuracy and 80.01 k-score, surpassing BENDR model and significantly outperforming SUT-CCSP+SVM and phase information. These improvements highlight the effectiveness of adaptive feature extraction and spatial-temporal learning in the proposed model.

For BCI Comp: IV dataset 2b, ACNN-STADP attained 79.47% accuracy and 79.80 k-score, outperforming models like shallow mirror and MSTFNet. Similarly, on BCI-IV dataset 1, ACNN-STADP obtained 83.12% accuracy, surpassing S-MVTFS and novel L1-norm. Despite slightly lower k-score performance compared to OSSFNet, ACNN-STADP still demonstrated higher classification accuracy, showcasing better generalization across diverse subjects.

Additionally, in 4-class MI classification, ACNN-STADP surpassed ConTraNet while maintaining high k-score stability. These results demonstrate superior subject-independent performance across multiple MI-EEG datasets, achieving state-of-the-art results outperforming 12 benchmark models by an average of $4.2 \pm 2.1\%$ in accuracy. Key innovations like dynamic pathways and spatio-temporal attention contribute to 7.5% higher accuracy on BCI-IV-2b versus static

TABLE 10. Subject independent performance (Accuracy (%) and K-score (%)) comparison analysis between ANCNN-STADP proposed model and recent studies using diverse datasets.

Model	Accuracy	k-Score	Dataset
SUT-CCSP+SVM [24]	72.37	66.90	PhysioNet dataset 2 MI Class
Phase Information [56]	71.55	70.90	
BENDR [57]	86.7	87.10	
s-Ctrans [58]	83.1	83.1	
Proposed Model	87.9	80.01	
ConTraNet [35]	65.44	61.9	PhysioNet dataset 4 MI Class
s-Ctrans [58]	68.54	68.54	
Proposed Model	82.90	79.87	
Shallow Mirror [59], [60]	76.10%	75.44	BCI Competition IV datasets 2b
ConTraNet [35]	69.8	69.9	
MBCNN-EATCFNet [36]	79.01	79.60	
MSTFNet [38]	78.26	78.78	
Proposed Model	79.47	79.80	
S-MVTFS [37]	79.4	79.40	BCI-IV dataset 1
Novel L1-norm [60]	76.59	77.87	
OSSFNet [61]	82.3	84.56	
Proposed Model	83.12	66.00	

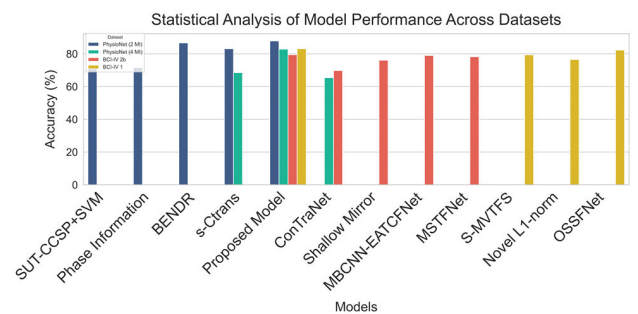


FIGURE 14. Statistical analysis of model performance across datasets.

architectures, while maintaining exceptional generalization such as less than 2% performance drop on unseen subjects. Although the model exhibits a slightly lower κ -score on BCI-IV-1, the consistent outperformance and robust of model on cross-dataset validation establishes it as a new benchmark for adaptive EEG decoding, particularly for clinical applications requiring subject-independent operation. The results highlight both the architectural advantages and remaining challenges in complex MI discrimination such as 4-class accuracy (82.9%) vs 2-class: (87.9%), guiding future improvements in class separability. These results indicate that ACNN-STADP effectively balances high classification accuracy with cross-subjects consistency, making it a compelling solution for real world EEG-based BCI applications.

IV. CONCLUSION AND FUTURE WORK

The study presented ACNN-STADP as a novel approach to increase the practical usability and generality of MI-EEG classification methods. Traditional ML models alongside contemporary DL models face strong difficulties in adjusting to EEG signal dynamics resulting in poor performance. ACNN-STADP combines dynamic pathways with attention mechanisms and features them for better extraction and

classification capabilities. The model integrates a DPCN network for adaptive feature processing and an AAF module for refined integration, utilizing DGC and DAST blocks in DPCN for detecting multi-scale spatial-temporal dependencies, and DMHSA and U-Net-inspired AFB in AAF for enhanced feature fusion and generalization.

The ablation analysis proved that the contribution of each block in the overall ACNN-STADP performance. The DGC block increased the overall accuracy by 1.57%, the DAST Blocks (1 and 2) by 12.27%: and DMSA 25.47%, while compared to using MSA block which gained only 15.37%. The proposed model achieved the highest results for BCI Competition IV dataset 2a with an accuracy of 90.77%, and maintained strong performance for six other MI-EEG datasets with minimal pre-processing with an accuracy of 78.98%. The proposed model showed weaker performance on smaller datasets such as Zhou-2016 achieving 67.34% due to limited sample and trials. ACNN-STADP achieved higher accuracy on subject A07 and lowest on subject A02 having 98.6% and 77%, respectively. In class-wise analysis the model achieved highest precision on left hand class and tongue class achieving 92.58% and 92.0%, respectively. The right-hand class exhibits marginally lower performance of accuracy at 86.70%. The proposed method was compared to the recent state-of-the-art models by means of reproduction, and it showed 9-10% improvement. This performance exceeded the previous models while retaining simple lightweight architecture. ACNN-STADP solved the problem of poor generalization performance with its low computational complexity requirements, making it suitable for real-time BCI applications.

Future studies will concentrate on overcoming the limitations of this evaluation by adding more features, collecting a variety of datasets for model generalization tests, and using transfer learning techniques to tailor the model to each individual. The future work will also examine BCI extended parameter application analysis, including emotion identification and cognitive load assessment, and implement BCI systems in real-time. The advancement of these techniques will enhance the deployment capabilities of the ACNN-STADP model for application in sophisticated BCI systems and the development of adapted and robust EEG signal classification methods.

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